



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G3 SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: Unitra

Prepared By: Justine C. Vilocura

Date of Submission: 02/06/2026

Version: 2

Table of Contents

- 1. Introduction.....3
 - 1.1. Purpose..... 3
 - 1.2. Scope..... 3
 - 1.3. Definitions, Acronyms, and Abbreviations..... 3
- 2. Overall Description.....3
 - 2.1. System Perspective..... 3
 - 2.2. User Classes and Characteristics.....3
 - 2.3. Operating Environment..... 3
 - 2.4. Assumptions and Dependencies..... 3
- 3. System Features and Functional Requirements.....3
 - 3.1. Feature 1:.....3
 - 3.2. Feature 2:.....3
- 4. Non-Functional Requirements..... 3
- 5. System Models (Diagrams)..... 4
 - 5.1. ERD..... 4
 - 5.2. Use Case Diagram..... 4
 - 5.3. Activity Diagram.....4
 - 5.4. Class Diagram.....4
 - 5.5. Sequence Diagram.....4
- 6. Appendices.....4

1. Introduction

1.1. Purpose

The purpose of this document is to describe the functional and non-functional requirements of **Unitra: SME Resource Sharing & Collaboration Network**. It is intended for project stakeholders, developers, instructors, and evaluators to understand the system's scope, features, and operational requirements.

1.2. Scope

Unitra is a web-based platform that enables Small and Medium Enterprises (SMEs) to create business profiles, share and request resources, exchange services, and communicate securely. The system focuses on collaboration and resource optimization and does not include online payments, advanced analytics, or mobile-native applications.

1.3. Definitions, Acronyms, and Abbreviations

- **SME:** Small and Medium Enterprise.
- **MVP:** Minimum Viable Product (the initial functional prototype).
- **UAT:** User Acceptance Testing.

2. Overall Description

2.1. System Perspective

The system operates as a web-based collaborative network intended to replace informal, unverified communication methods (like social media groups) with a structured, professional business environment.

2.2. User Classes and Characteristics

- **SME Owners/Operators:** Primary users who manage business profiles and list/request resources.
- **System Administrators:** Responsible for managing the platform, verifying users, and overseeing community boards.

2.3. Operating Environment

The system will be a web application built using **Django (Python)** for the backend and **React.js** for the frontend. It will utilize **MySQL or PostgreSQL** for data storage and be hosted in a **Cloud** environment accessible via standard web browsers on desktops and mobile devices.

2.4. Assumptions and Dependencies

- **Assumption:** SMEs are willing to share equipment and services with local competitors or partners.
- **Dependency:** Stable internet connectivity is required for the real-time messaging and marketplace updates.

- **Dependency:** Project success depends on achieving at least 75% positive feedback during usability testing.

3. System Features and Functional Requirements

3.1. Feature 1: SME Profile Management

Description: Allows SMEs to register, create, and manage their business profiles within the platform.

Functional Requirements:

- SMEs shall be able to register and log in to the system
- SMEs shall be able to create and update business profile information
- The system shall store profile data securely in the database

3.2. Feature 2: Resource Sharing & Service Exchange

Description: Enables SMEs to list, request, and exchange resources or services with other SMEs.

Functional Requirements:

- SMEs shall be able to list available resources or services
- SMEs shall be able to request or reserve shared resources
- The system shall prevent double booking of the same
- The system shall update transaction status (pending, approved, declined) in real time

4. Non-Functional Requirements

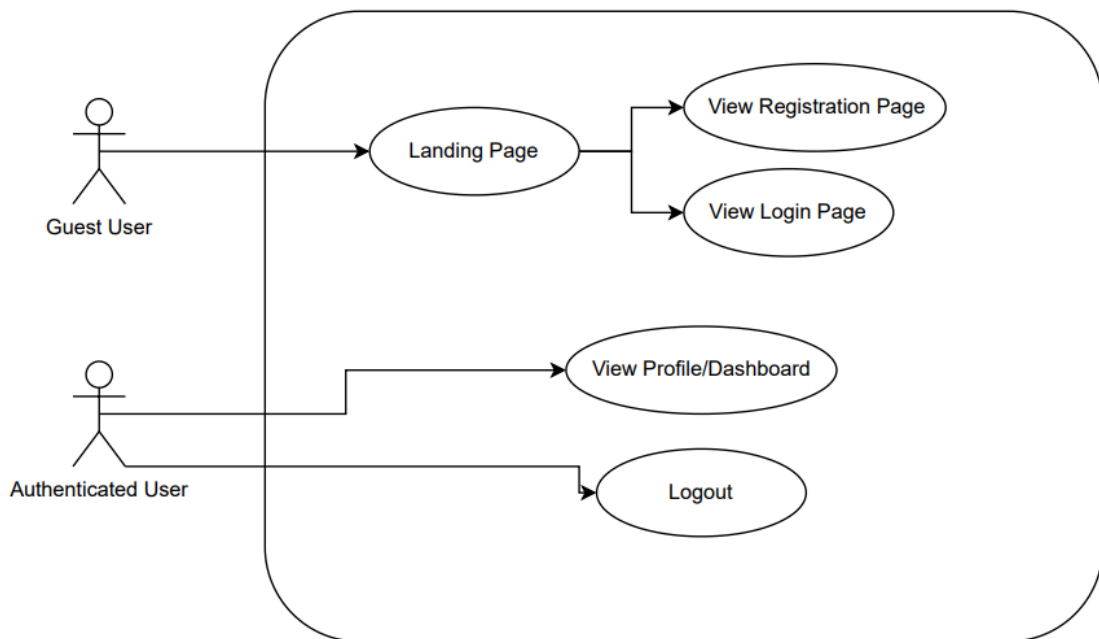
- **Security:** The system shall use secure logins, hashed passwords, and HTTPS to protect sensitive business data.
- **Usability:** The interface must be clean and intuitive, ensuring at least 75% positive feedback from non-tech-savvy users.
- **Reliability:** The system shall utilize cloud-based hosting to ensure data backups and high availability.
- **Performance:** The database shall be optimized to handle a growing number of SMEs without significant latency.

5. System Models (Diagrams)

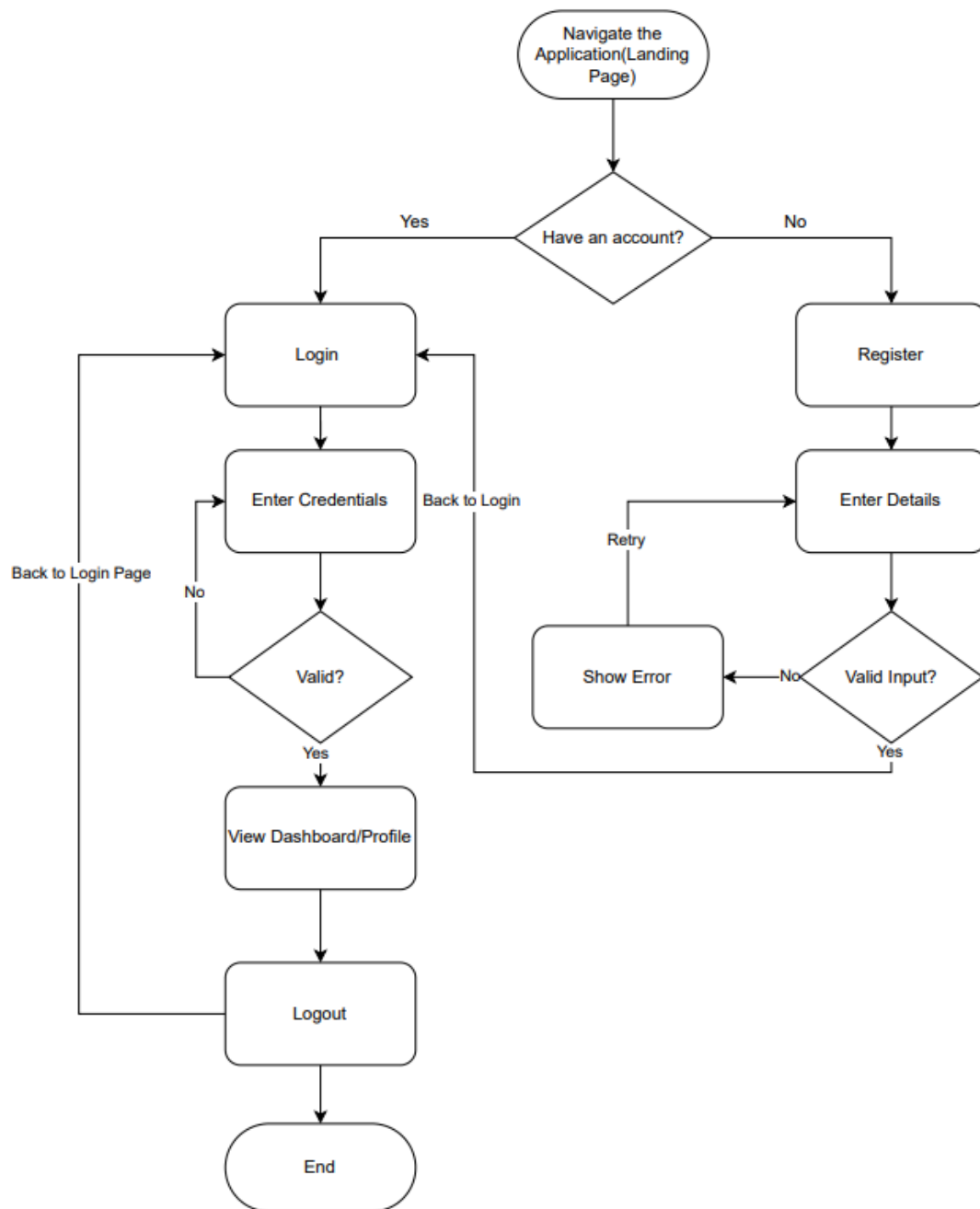
5.1. ERD

Users	
Field Name	<u>Data Type</u>
user_id	BIGINT (PK)
username	VARCHAR(50)
email	VARCHAR(100)
password	VARCHAR(255)
role	VARCHAR(20)

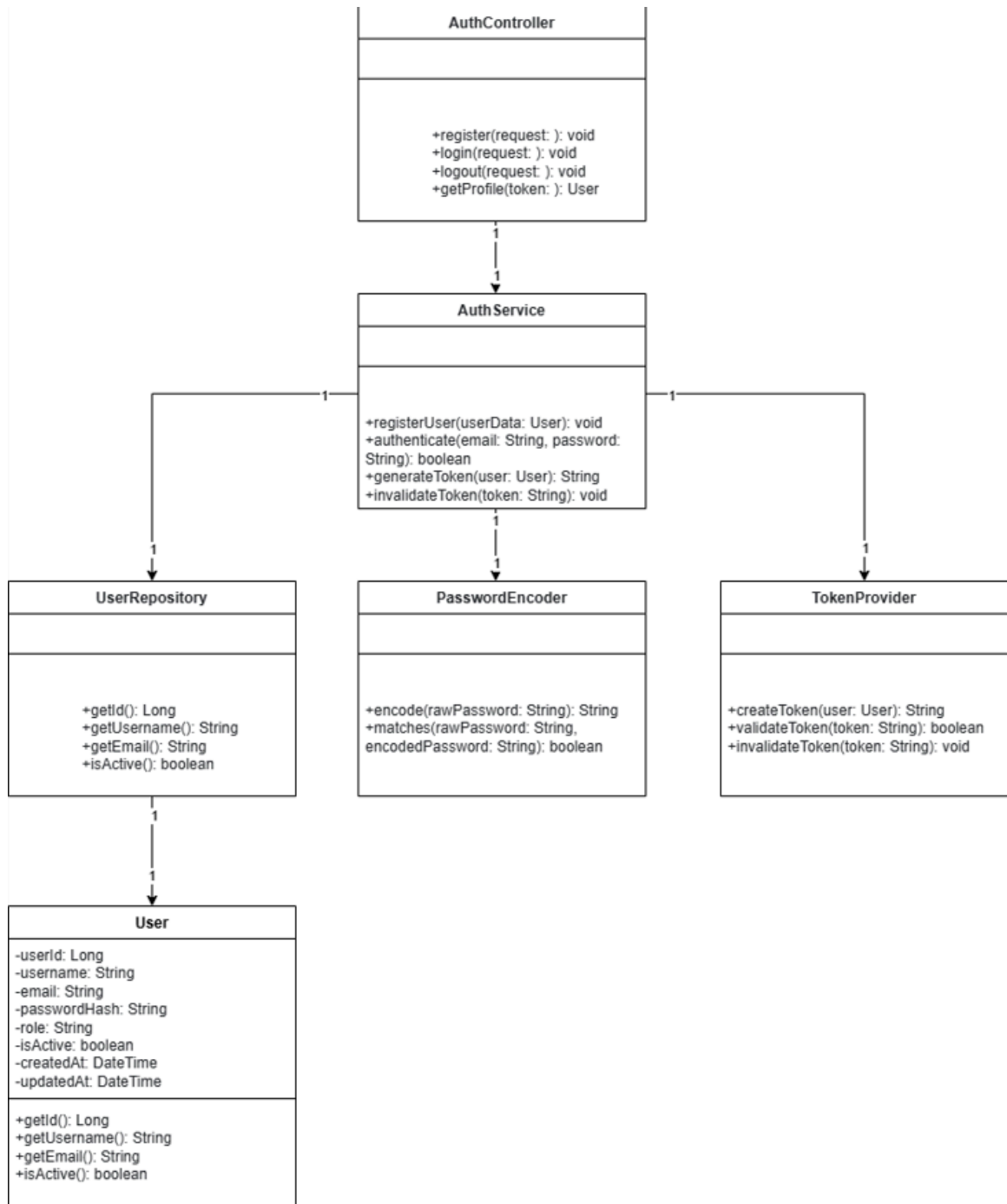
5.2. Use Case Diagram



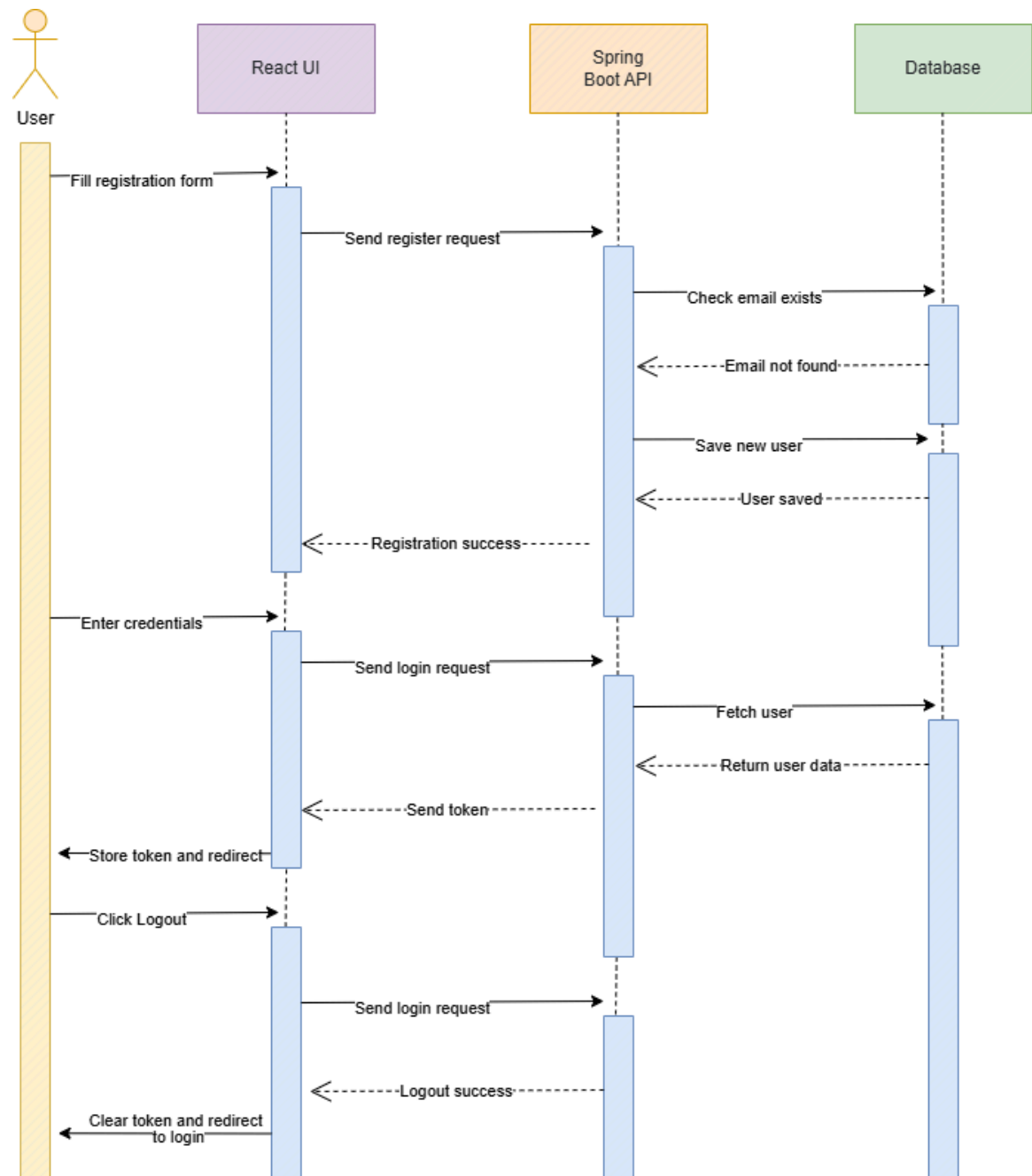
5.3. Activity Diagram



5.4. Class Diagram

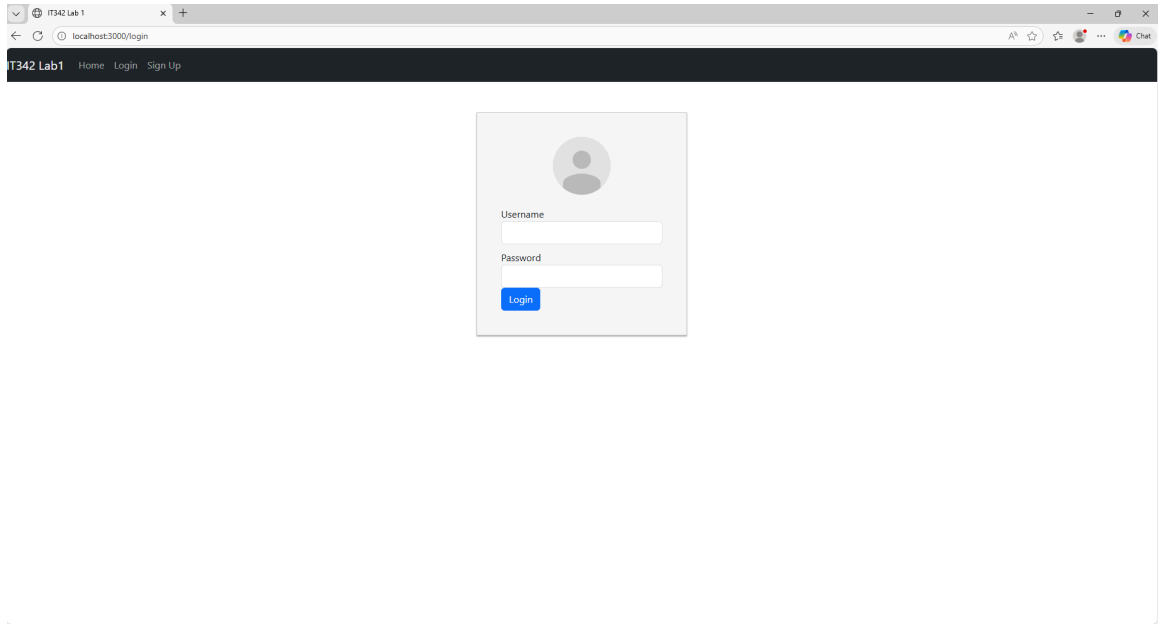


5.5. Sequence Diagram

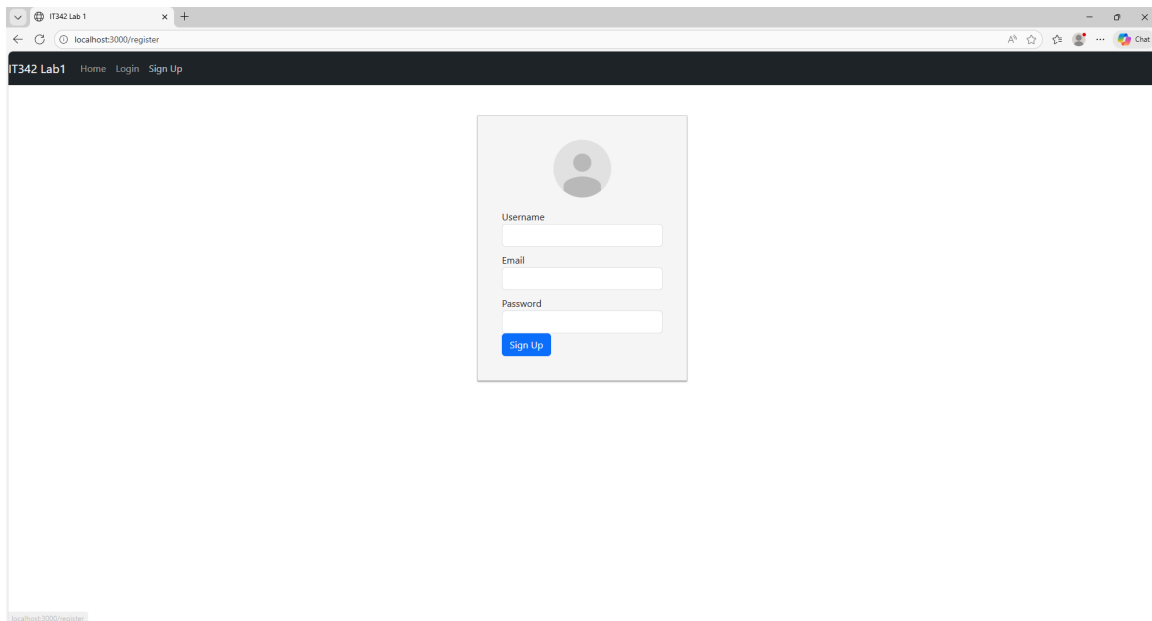


6. Screenshots of the Web UI

Login:



Register:



Dashboard:

7. Appendices

Appendix A: Use Case Summary

This appendix provides a summary of key use cases supported by the Unitra system:

- SME Registration and Account Verification
- Business Profile Creation and Management
- Resource Listing and Reservation
- Service Exchange and Bartering Requests
- Direct Messaging and Group Discussions
- Community Announcement Posting

Appendix B: Sample Transaction Flow

1. An SME lists a resource as available.
2. Another SME submits a request to use the resource.
3. The system checks availability and booking conflicts.
4. The resource owner approves or declines the request.
5. The system updates the transaction status and records the exchange in the database.

Appendix C: Assumptions and Constraints

- The system will initially support web-based access only.
- Secure payment processing is excluded from the current scope.
- User adoption depends on effective onboarding and training.
- System performance depends on cloud hosting reliability and internet connectivity.

Appendix D: References

- Patrucco A.S., Trabucchi D., Buganza T., Muzellec L., and Ronteau S.(2024). *Technology-enabled multi-sided platforms in B2B relationships: A critical analysis and directions for future research*
- Levina N. (2005). *Collaborating on Multiparty Information Systems Development Projects: A Collective Reflection-in-Action View.*
- Br Tarigan KE, Sherine S, Muda I and Gusnardi G. (2025). *Collaboration Systems and Social Business: Important Pillars in the Digital Era.*